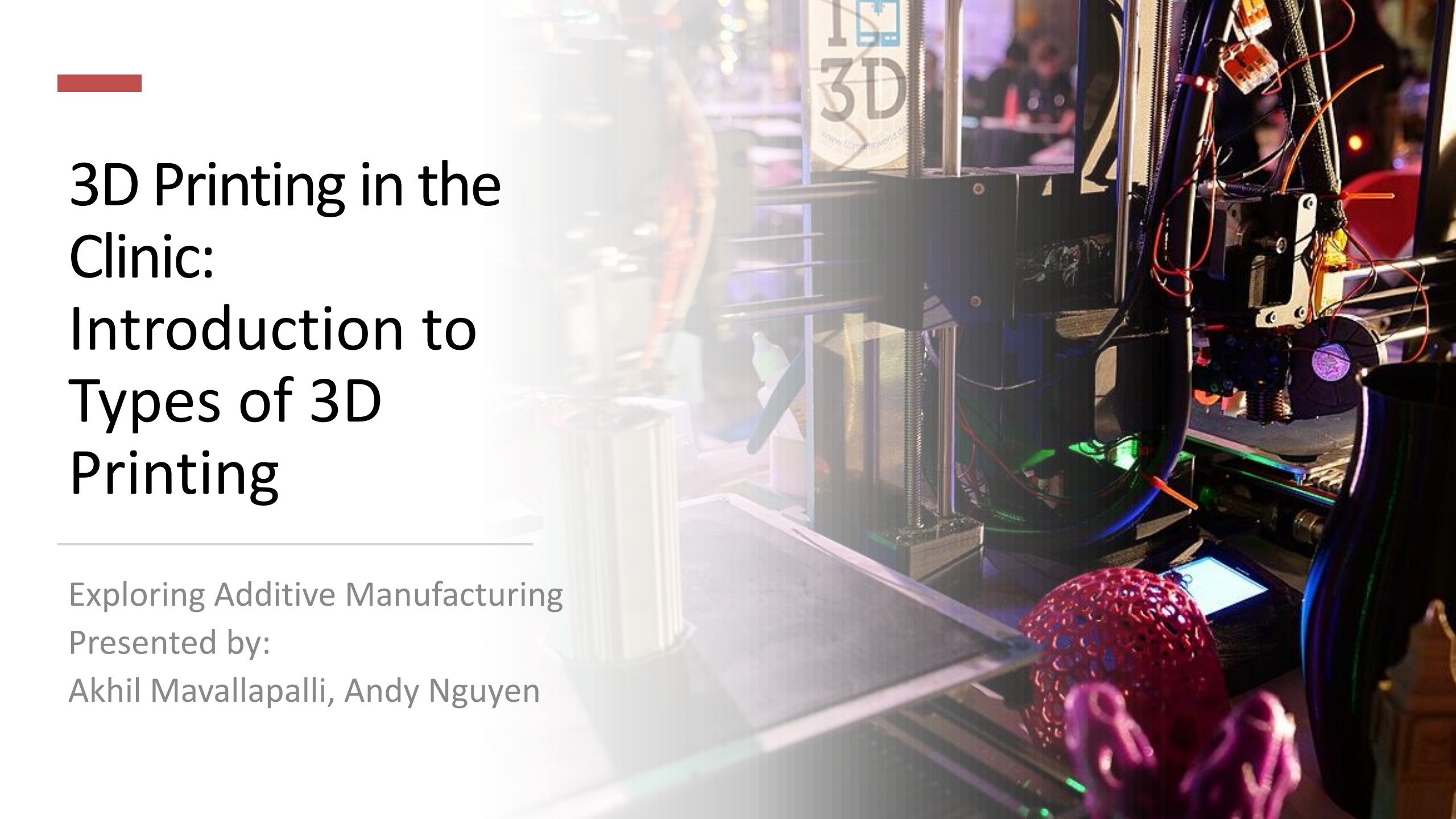


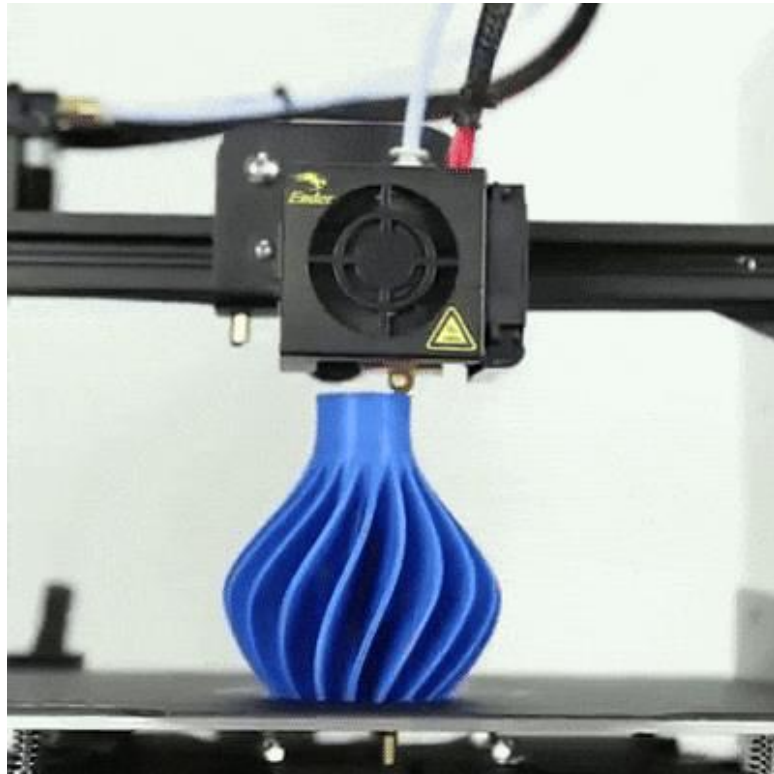


3D Printing in the Clinic: Introduction to Types of 3D Printing

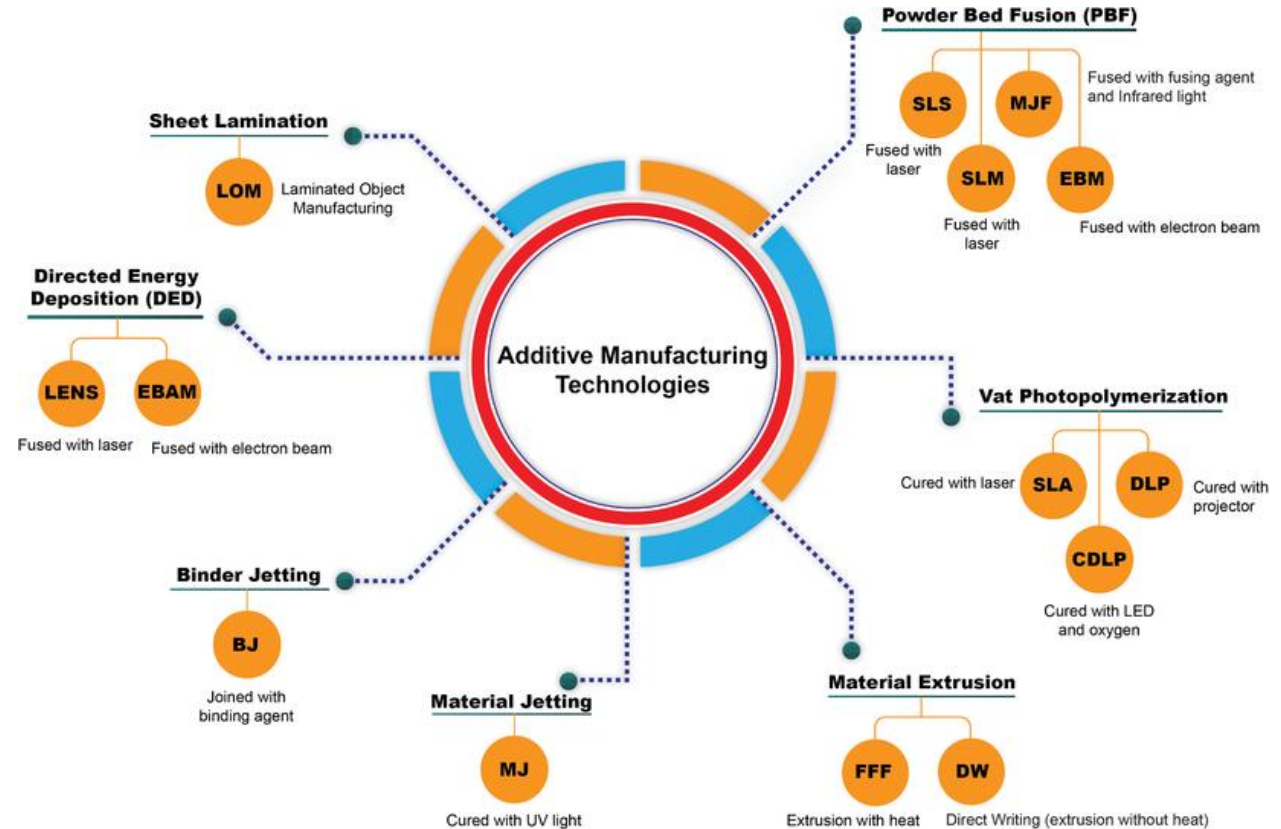
Exploring Additive Manufacturing
Presented by:
Akhil Mavallapalli, Andy Nguyen

Introduction – What is 3D Printing?

- Additive manufacturing builds parts layer by layer
- Enables rapid prototyping, customization, and reduced waste
- Evolved from 1980s stereolithography to bioprinting
- Key takeaway: Different techniques serve different materials and needs



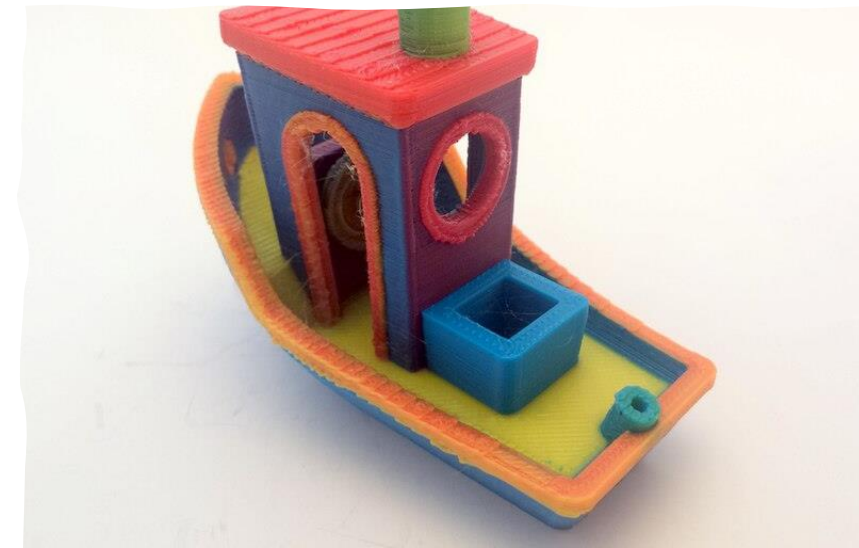
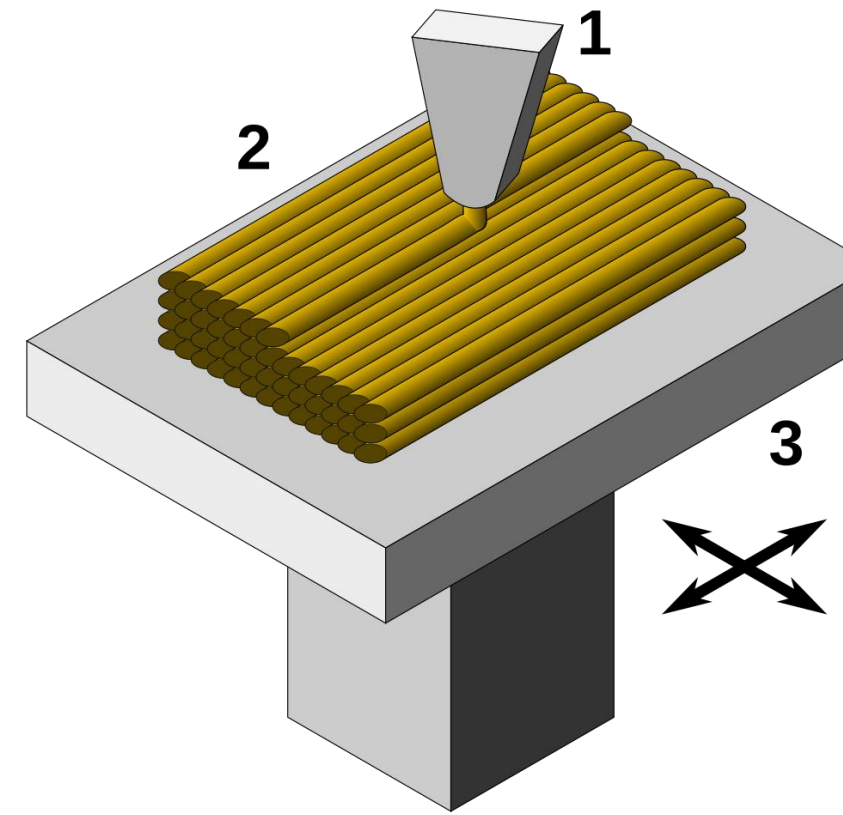
Major Categories of 3D Printing



- Polymer-based: FDM, SLA, SLS – used in prototyping and medicine
- Metal-based: SLM, EBM – implants
- Composite/Ceramic: Binder Jetting – dental and tooling
- Bioprinting: Extrusion, Inkjet – regenerative medicine

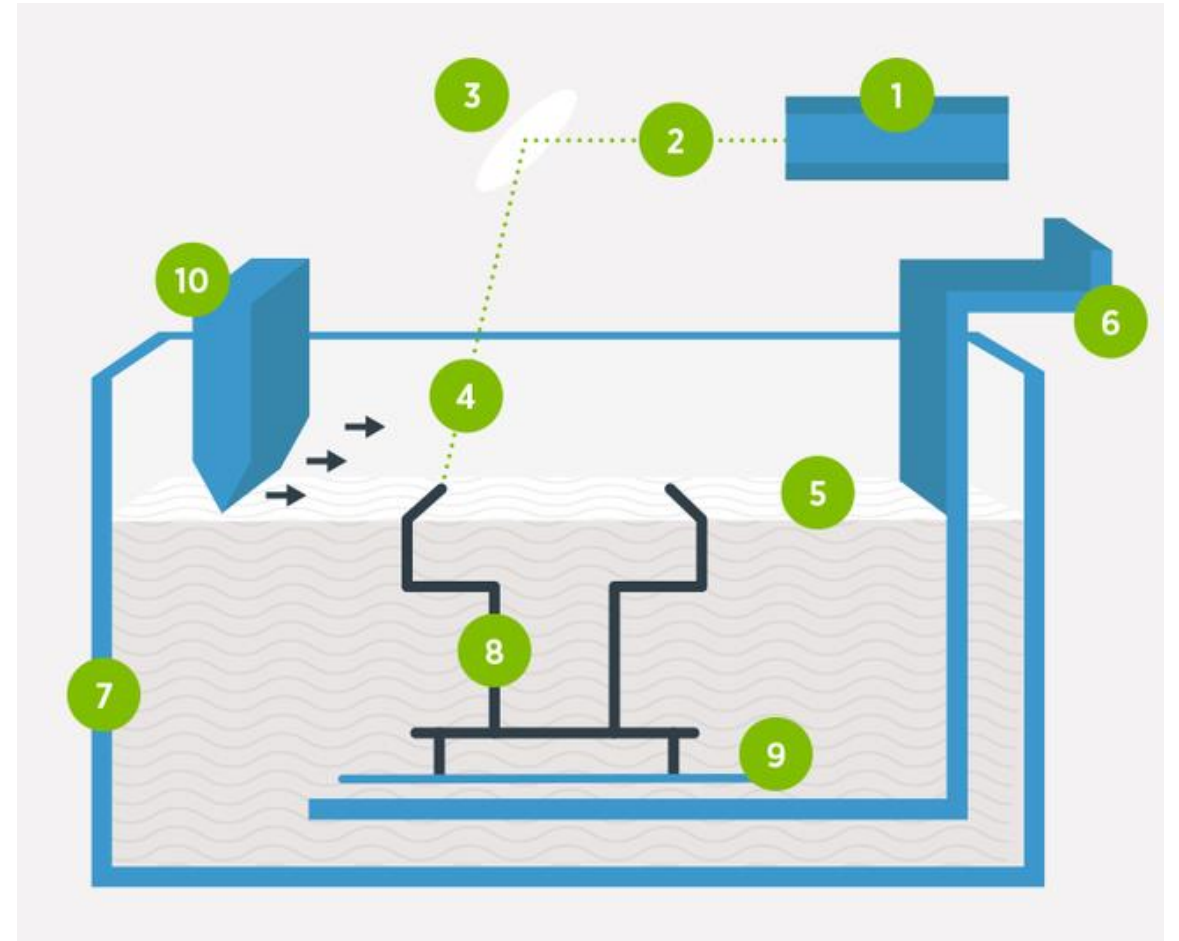
Fused Deposition Modeling (FDM)

- Heated filament extruded layer-by-layer
- PLA, ABS, PETG, TPU
- Medical: Custom pediatric prosthetic sockets, orthotics
- Nonmedical: Drone airframes, rapid prototypes
- **Pros:** Low cost, easy to use, wide material choice, fast prototyping
- **Cons:** Rough surface, weaker layer bonding, limited accuracy



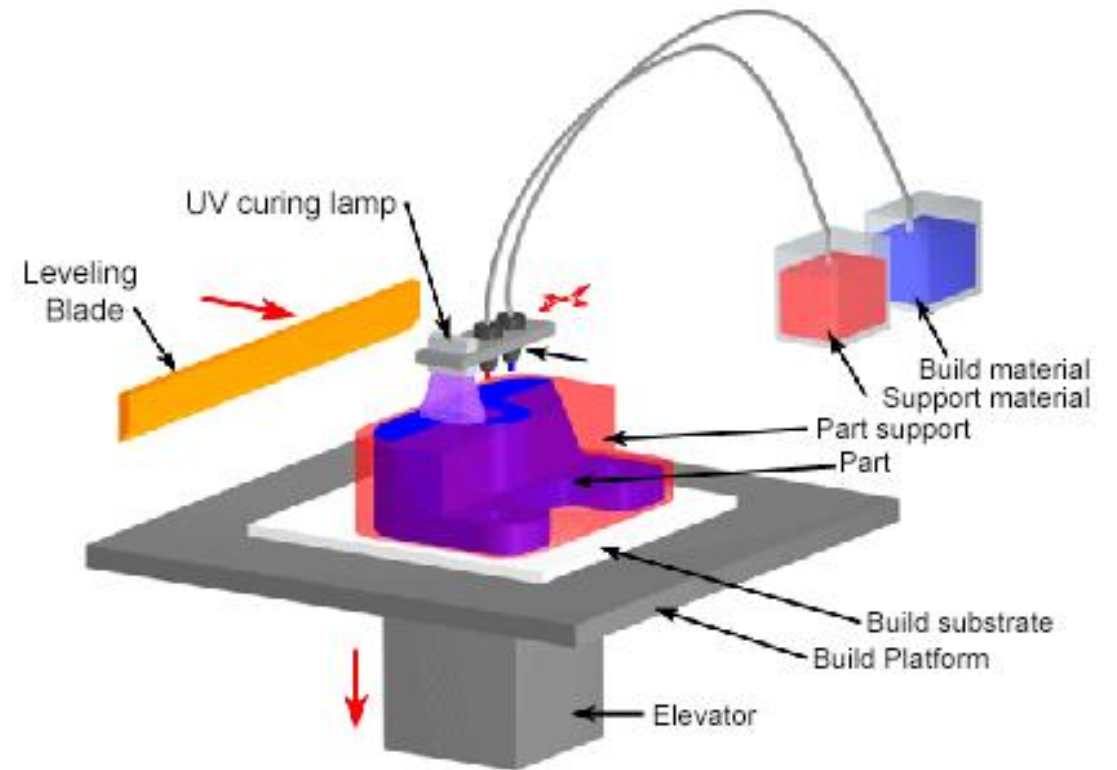
Stereolithography (SLA)

- Laser or LCD light cures resin
- Photopolymer resins
- Medical: Dental crowns, surgical guides
- Nonmedical: Electronics casings, figurines
- **Pros:** High precision, smooth surface finish, good for small parts.
- **Cons:** Brittle materials, resin handling mess, post-curing required.



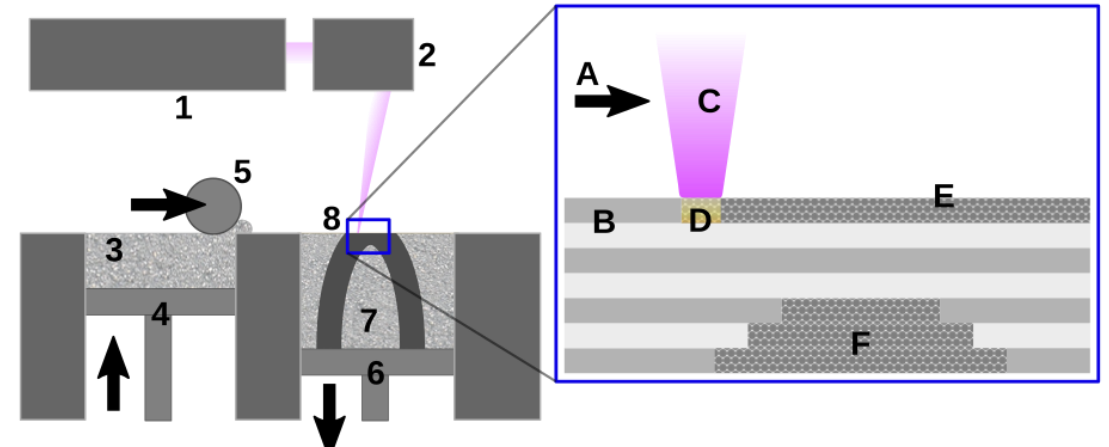
Material Jetting (PolyJet)

- Photopolymer droplets cured by UV
- Medical: Multicolor anatomical hearts
- Nonmedical: Product prototypes
- **Pros:** Multi-material and full color, excellent surface finish, high accuracy
- **Cons:** High material cost, limited durability, slower build speed



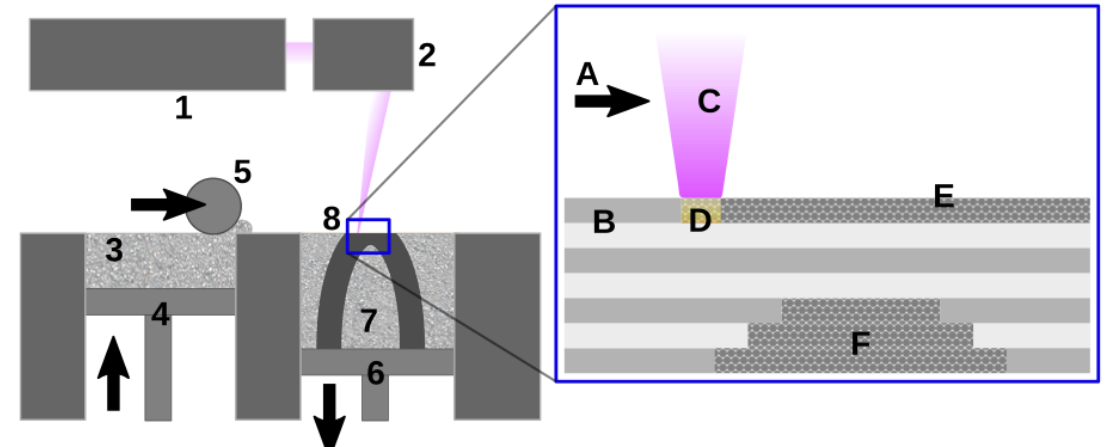
Selective Laser Sintering (SLS)

- Laser fuses polymer powder without supports
- Nylon12, composites
- Medical: Prosthetic sockets, implant prototypes
- Nonmedical: Automotive clips, housings
- **Pros:** Strong parts, no supports, good for complex shapes.
- **Cons:** Rough surface, expensive machines, slower cooling time.



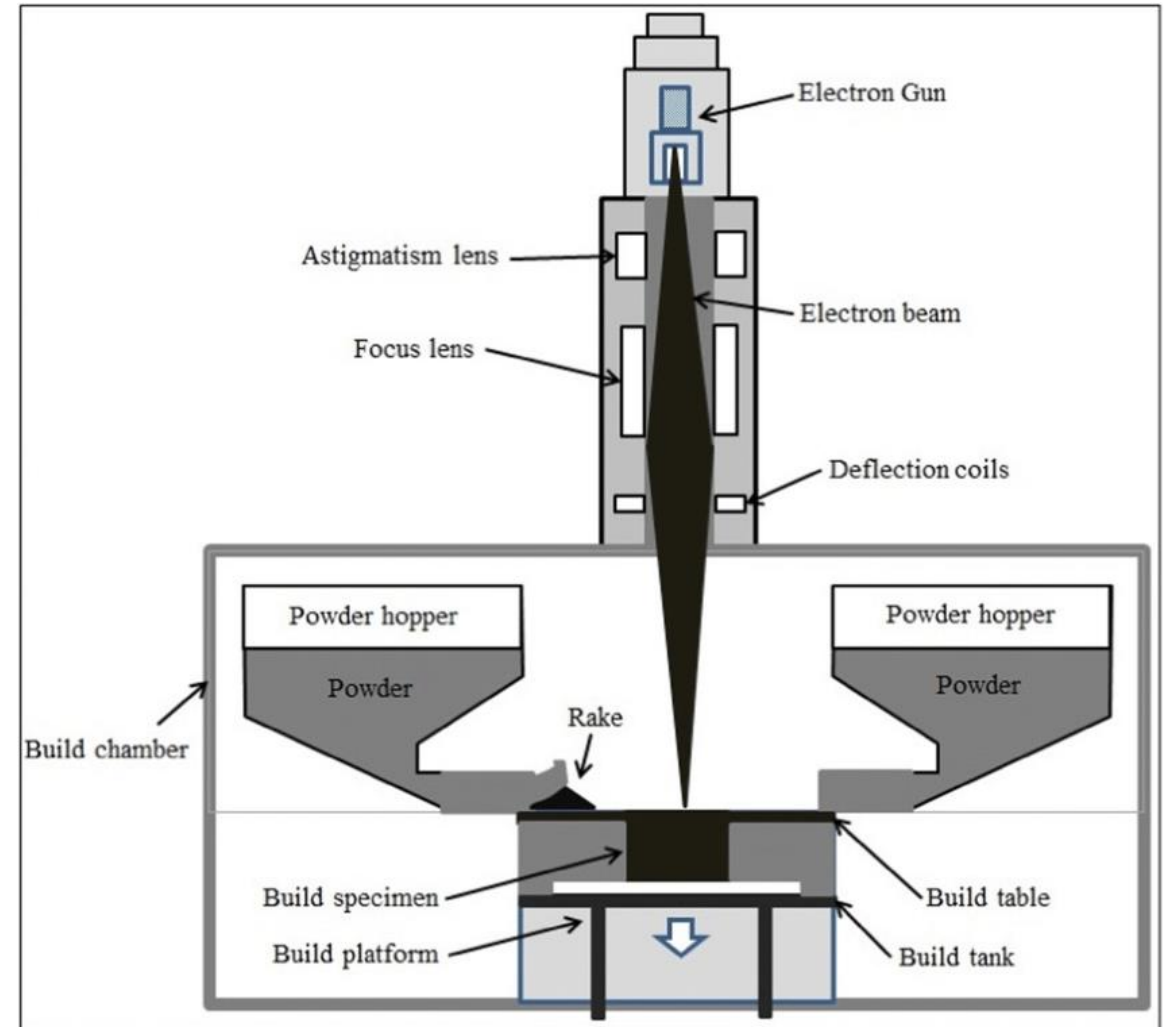
Selective Laser Melting (SLM) / DMLS

- Laser fully melts metal powder
- Titanium, cobaltchrome, stainless steel
- Medical: Spinal cages, hip implants
- Nonmedical: Aerospace turbine brackets
- **Pros:** Fully dense metal parts, excellent strength, high detail.
- **Cons:** Very expensive, high energy use, internal stresses.



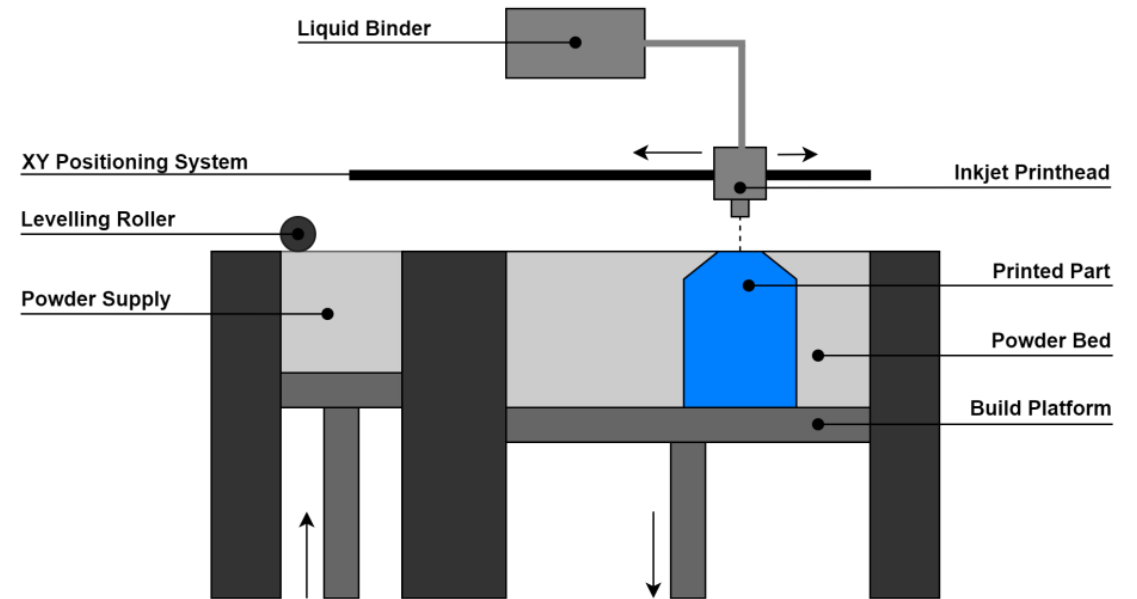
Electron Beam Melting (EBM)

- Electron beam fuses powder in vacuum
- Titanium alloys
- Medical: Orthopedic implants
- Nonmedical: Satellite thruster mounts
- **Pros:** Great for titanium, low residual stress, vacuum environment.
- **Cons:** Coarse surface finish, limited materials, expensive setup.



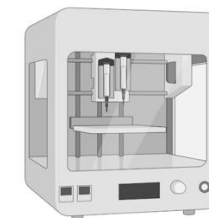
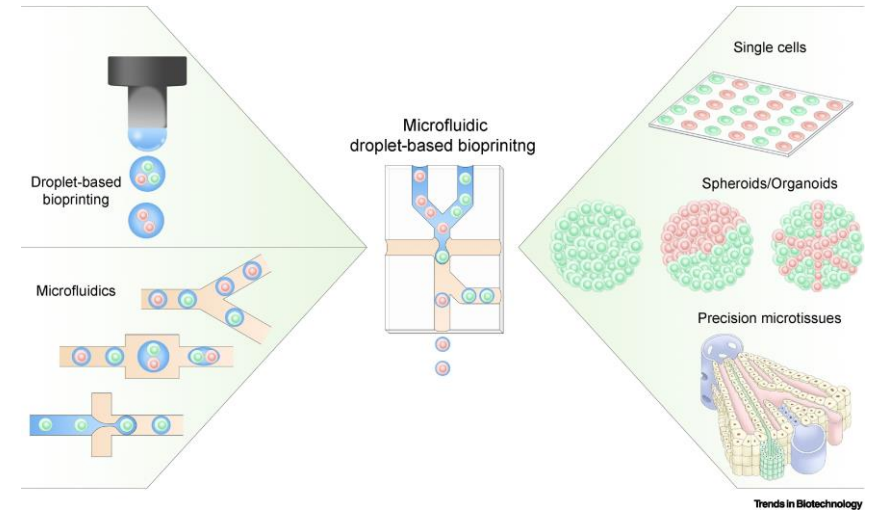
Binder Jetting

- Binder deposited on powder bed; later sintered
- Metals, ceramics
- Medical: Bone models for planning
- Nonmedical: Sand molds for foundry use
- **Pros:** Fast printing, scalable, low cost per part.
- **Cons:** Requires sintering, lower strength, limited precision.

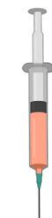


Bioprinting

- Deposition of bioinks (cells + hydrogels)
- Medical: Cartilage, skin scaffolds
- Nonmedical: Biomimetic materials, food gels
- **Pros:** Enables tissue engineering, customizable biological structures, huge research potential.
- **Cons:** Complex process, expensive, low long-term durability.



Bioprinter



Bioink

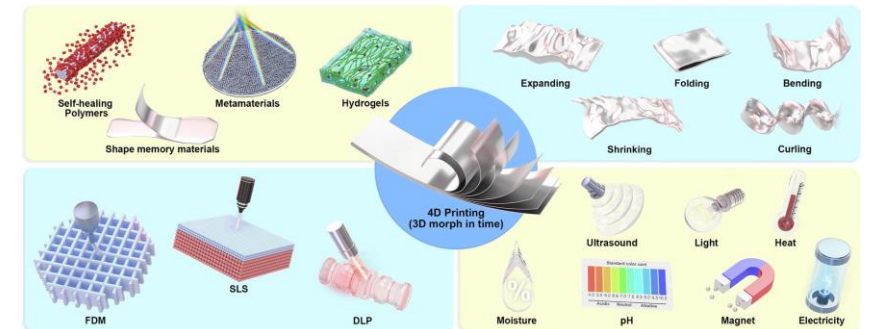
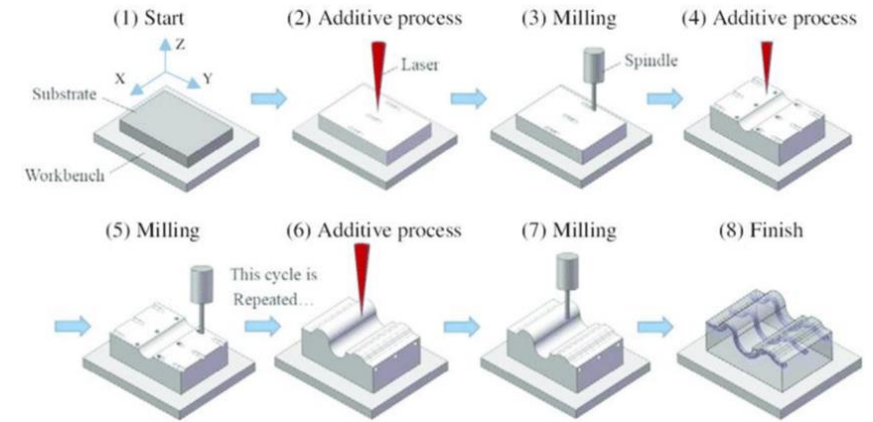


Cells Ink Additives ECM Components

Bioink Elements

Emerging Directions

- 4D printing – shapechanging materials
- Hybrid printing – combining additive + subtractive
- Recycled feedstocks – sustainable future
- **Pros:** Smart materials, high precision, environmentally friendly
- **Cons:** Experimental, costly equipment, limited commercial use (so far)



Comparative Summary Table

FDM – Orthotics / Drones

Pros: Cheap, easy, versatile

Cons: Rough surfaces, weaker layer adhesion

SLA – Surgical Guides / Figurines

Pros: Very high detail, smooth finish

Cons: Brittle, requires messy post-processing

SLS – Prosthetics / Automotive

Pros: Strong parts, no supports needed

Cons: Grainy surface, expensive machines

SLM – Implants / Jet Parts

Pros: Fully dense metals, excellent strength

Cons: Very high cost, residual stresses

Binder Jetting – Bone Models / Molds

Pros: Fast + scalable, low cost per part

Cons: Needs sintering, lower final strength

PolyJet – Organs / Product Prototypes

Pros: Multi-material, full color, top detail

Cons: Expensive resins, parts not durable

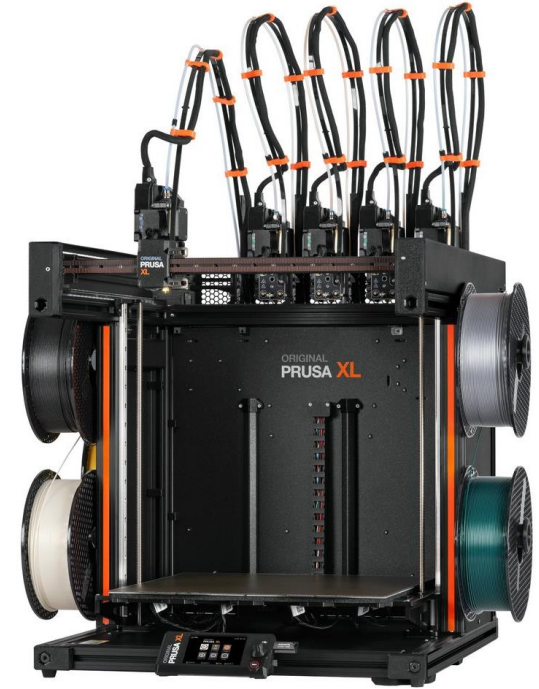
Bioprinting – Tissue / Food Gels

Pros: Enables tissue engineering, customizable biology

Cons: Experimental, limited long-term durability

Bakken Medical Devices Center Offerings

- Stratasys J850 Polyjet printer
- 3x Prusa MK4 printers
- 2x Prusa Mini+ printers
- Prusa XL



Q&A / Closing

- Thank you to BMDC and the Minneapolis VA RECOVER for helping put together this presentation



Earl E. Bakken
Medical Devices Center

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